

SUPPLIER QUALITY RATING SYSTEM (SQRS) AUDIT

Monthly Vendor Evaluation Report — Day 08 | Quality Engineering Challenge

Report Ref:	SQRS-BFL-2026-07	Audit Month:	July 2026
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Component:	Precision Helical Gears	Target Limit:	< 1,000 PPM Defect Rate

1. AUDIT BACKGROUND & KEY PRIORITIES

In our transmission line assembly, overall reliability relies heavily on the quality of incoming gear components. During July 2026, we audited Bharat Forge Limited to assess their monthly supply of precision helical gears. This rating framework helps us catch quality gaps early, avoid line stoppage due to delivery delays, and build a practical zero-defect workflow with our tier-1 vendors.

Core Operational Focus Areas:

- Defect Minimization (PPM):** Keeping rejected parts as low as possible during receiving checks.
- Delivery Timeliness (OTD):** Ensuring shipments arrive strictly on schedule so the assembly line never sits idle.
- Drawing Specification Compliance:** Making sure every gear strictly meets engineered micro-tolerances and surface finish requirements.
- Responsiveness:** Quick turnaround on quality queries and submitting effective 8D root-cause reports when issues pop up.

2. PARAMETER CALCULATION & FORMULA BREAKDOWN

Below is the step-by-step scoring breakdown showing exact evaluation formulas alongside actual operational figures:

Evaluation Metric	Formula Used (Mathematical Rule)	Monthly Data & Calculation	Score Earned
Quality Rating (PPM) (40% Weightage)	$PPM = (Rejections / Total Received) \times 10^6$ $Score = 40 \times [1 - (Rejections / Total)]$	Total Received = 50,000 units Total Rejections = 50 units PPM = 1,000 PPM Score = $40 \times [1 - (50 / 50,000)]$	39.96 / 40
Delivery Rating (OTD) (30% Weightage)	$Delivery Score = 30 \times (On-Time Batches / Total Scheduled)$	Scheduled Batches = 35 On-Time Batches = 35 Score = $30 \times (35 / 35)$	30.00 / 30
System Audit Score (15% Weightage)	$Audit Score = 15 \times (Audit Compliance \% / 100)$	Shopfloor Audit = 90% Score = $15 \times (90 / 100)$	13.50 / 15
8D Responsiveness (15% Weightage)	$Response Score = 15 \times (Turnaround Compliance Factor)$	Requirement: < 48 Hours Actual: Submitted in 24 Hours Score = 15×1.00	15.00 / 15

3. FINAL SQRS AUDIT SCORECARD

Evaluation Parameter	Weightage	Target Benchmark	Actual Result	Final Earned Score
Quality Performance (PPM)	40%	< 1,000 PPM	1,000 PPM (50 Rejections)	39.96 / 40
Delivery Performance (OTD)	30%	100% On-Time	35 / 35 Batches	30.00 / 30
Process System Audit	15%	> 85% Compliance	90% Audit Rating	13.50 / 15
8D Issue Turnaround	15%	< 48 Hours	Submitted in 24 Hours	15.00 / 15
TOTAL OVERALL SCORE	100%	Grade A (> 90%)	GRADE A PREFERRED	98.46 / 100

4. NEXT STEPS & VENDOR IMPROVEMENT PLAN

Scoring **98.46%** keeps Bharat Forge in our **Grade A Preferred Vendor** category. To push our quality standards further towards a zero-defect target, we are putting three main plans into action:

- **Direct Vendor Integration (Early Involvement):** We are involving Bharat Forge application engineers directly in our plant design reviews. Catching tooling and tolerance issues during design saves both teams time and prevents unexpected floor rejections.
- **PDCA Action Plan:**
 - *Plan:* Work together to reduce incoming defects down from 1,000 PPM to under 500 PPM by next quarter.
 - *Do:* Run regular spot-checks at receiving and verify critical pitch measurements.
 - *Check:* Review monthly defect Pareto charts and double-check shopfloor capability stats (Cp/Cpk).
 - *Act:* Standardize working steps across both teams once a corrective fix proves successful.
- **Mistake-Proofing (Poka-Yoke):** Bharat Forge has installed an optical pitch sensor on their gear hobbing machine to check tooth geometry before packing, stopping defective gears right at their plant.

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